GA-Driven Recurrent Modeling for Indonesian Banking Stocks Forecasting with Transformer and Recurrent-Variant Extensions

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***Abstract*—This paper studies daily closing-price forecasting for major Indonesian banking stocks using recurrent sequence models, macroeconomic regressors, and statistical validation. The primary ablation compares LSTM and BiLSTM under two feature settings (`Without_Macro` and `With_Macro`) and two optimization modes (default vs. budgeted tuning via Genetic Algorithm), resulting in eight experimental cases per bank. To improve rigor, we run repeated multi-seed evaluations across seven banks and apply Wilcoxon signed-rank, Diebold-Mariano, and SHAP-based analyses. Empirically, BiLSTM-based configurations dominate most banks, with best-case RMSE improvements over the baseline ranging from 4.91% to 35.66% for six out of seven banks. As an extension, we further evaluate Informer/Autoformer Transformer models and recurrent variants based on normalization, convolution, and temporal attention. The Transformer extension shows strong aggregate performance, with `informer_data` achieving RMSE 401.02, MAE 325.11, and MAPE 5.94%, while recurrent variants do not outperform the original recurrent benchmark on average. These extension results provide promising directions for future statistically matched validation.**

***Keywords*—stock forecasting, LSTM, BiLSTM, Informer, Autoformer, macroeconomic features, genetic algorithm, Wilcoxon test, DM test, SHAP**

I. Introduction

Forecasting bank stock prices remains challenging due to non-stationarity, regime shifts, and sensitivity to macroeconomic conditions. In emerging markets such as Indonesia, banking-sector prices are influenced not only by internal price dynamics but also by external drivers including interest rates, inflation, exchange rates, and broad market movements.

Recent work in financial time series forecasting has adopted deep learning models, especially recurrent networks and Transformer-based architectures, to capture temporal dependencies. However, many studies still suffer from limited ablation depth and weak statistical validation. These gaps reduce confidence in whether reported improvements generalize across assets and random seeds.

This study develops a reproducible pipeline for Indonesian banking stocks that combines sequence modeling, macroeconomic regressors, repeated-seed evaluation, formal significance testing, and explainability analysis. A central question is whether feature engineering with macroeconomic variables (`With_Macro`) provides consistent gains over the baseline regime (`Without_Macro`), as often suggested in exogenous-variable forecasting literature [1], [2]. In addition, we add supplementary experiments with stronger architectures, especially Informer, Autoformer, and recurrent variants, to examine whether the recurrent benchmark can be extended. Model behavior is inspected with SHAP-based attributions where outputs are available [3].

The main contributions are as follows:

- A structured recurrent ablation design over model type (LSTM vs BiLSTM), feature regime (price-only vs price+macro), and optimization mode (off vs on).
- A multi-bank, multi-seed protocol for robust performance estimation and variance analysis.
- Statistical validation using Wilcoxon signed-rank and Diebold-Mariano tests to assess significance of recurrent forecasting improvements.
- An XAI layer based on SHAP to quantify global feature relevance across banks, feature regimes, and optimization settings.
- A supplementary comparison against Informer/Autoformer and normalization, convolutional, and attention-based recurrent variants.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing. Section IV presents the proposed methodology. Section V details the experimental setup. Section VI discusses the results and significance tests. Section VII concludes and outlines future directions.

II. Related Works

Literature on financial time-series forecasting spans classical statistical models, recurrent architectures, and increasingly, Transformer-based designs. LSTM-type models remain widely used in stock and index forecasting because they capture nonlinear temporal dependence with manageable model complexity. Recent IEEE studies continue to report competitive recurrent variants, including CNN-BiLSTM and attention-based designs [4], [5]. Meanwhile, Transformer architectures such as Temporal Fusion Transformer (TFT) [6], Informer [7], Autoformer [8], and TimeXer [1] have demonstrated strong multi-horizon performance. Informer targets efficient long-sequence forecasting through ProbSparse attention and sequence distilling, while Autoformer uses decomposition and auto-correlation mechanisms for trend-seasonal structure. These prior works motivate a supplementary comparison between the statistically validated recurrent benchmark and newer Transformer or recurrent-variant architectures.

Another key direction is forecasting with exogenous variables. Recent studies show that incorporating external drivers can improve prediction accuracy when those signals are informative and temporally aligned [2], [9]–[11]. This motivates our explicit comparison between `Without_Macro` and `With_Macro` settings.

Explainability is also becoming important in financial AI because predictive accuracy alone is insufficient for risk-sensitive decisions. SHAP provides a model-agnostic additive attribution framework and has been used in stock-market factor analysis to interpret model drivers [3], [12]–[14]. Recent IEEE studies also emphasize XAI-oriented analysis for financial forecasting workflows, including LRP- and dashboard-based explainability settings [15]–[18]. Compared with black-box reporting, SHAP-based summaries can reveal whether the dominant drivers are stable across assets, seeds, and modeling regimes.

Beyond architecture and features, methodological rigor remains a common weakness in applied financial forecasting papers. Many studies report single-run metrics without repeated-seed robustness checks or formal significance testing. This paper addresses that gap by combining controlled ablation, multi-seed evaluation, and paired statistical tests (Wilcoxon and Diebold-Mariano) in one reproducible pipeline for Indonesian banking stocks.

III. Dataset and Preprocessing

A. Dataset Description

This study focuses on seven Indonesian banking stocks: BBCA, BBNI, BBRI, BBTN, BDMN, BMRI, and BNGA. Daily market data include price and valuation-related variables, while macroeconomic variables include inflation-related indicators, policy rate, exchange rate, and market index features.

The target variable is the daily closing price. The analysis period follows a chronological split to avoid data leakage.

B. Feature Selection

The baseline feature set (`Without_Macro`) includes market micro and technical variables available in the bank-level dataset. The macro-augmented regime (`With_Macro`) extends this set with external variables:

- inflation rate and CPI-derived indicators,
- BI policy rate,
- USD/IDR exchange rate,
- IHSG index level and return proxy.

The final feature schema is fixed per regime before model training.

C. XAI-Oriented Feature Schema

To support consistent SHAP analysis across model families, feature names are kept identical across banks and seeds, and the same temporal windowing process is used in every run. The `Without_Macro` regime uses 10 market/valuation features, while `With_Macro` adds 6 external variables (total 16 features). This fixed schema allows direct aggregation of feature attribution statistics for original recurrent and Transformer outputs, and it keeps the recurrent-variant outputs compatible with future SHAP analysis.

D. Data Preprocessing

We apply chronological sorting, forward-fill for missing observations, and standard scaling for both features and target during model training. Sequence windows are then constructed with a fixed lookback length.

Time split protocol:

- Training: 2014-01-02 to 2019-12-31
- Validation: 2020-01-02 to 2021-12-31
- Testing: 2022-01-02 to 2024-12-30

This split reflects strict temporal validation and supports fair out-of-sample evaluation.

IV. Proposed Methodology

A. Overall Framework

The pipeline consists of five stages: data loading and scaling, sequence generation, model training, out-of-sample forecasting, and interpretability analysis. We evaluate multiple design factors using controlled ablation to isolate contribution from architecture, macro features, and optimization. Figure 1 summarizes the end-to-end workflow used in this study.

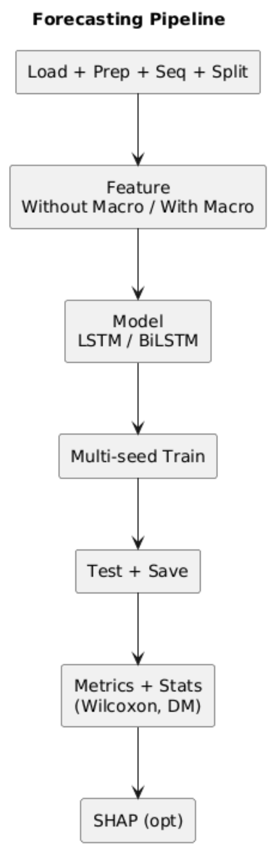

Fig. 1. Compact vertical pipeline overview used for the ablation and evaluation workflow.

B. Primary Recurrent Forecasting Framework

1) LSTM and BiLSTM Baselines: The original recurrent baselines use a single-layer recurrent encoder followed by a fully connected prediction head mapping the final hidden state to a scalar forecast. The shared architectural design ensures that performance differences are attributable to the directionality of temporal encoding rather than depth or capacity changes. The two variants are:

- **LSTM**: unidirectional temporal encoder with hidden size h, producing a single forward hidden state of dimension h.
- **BiLSTM**: bidirectional temporal encoder with hidden size h, concatenating forward and backward states to produce an output of dimension $2h$.

Both variants apply a dropout rate of 0.2 between the recurrent layer and the prediction head to mitigate overfitting. The prediction head consists of a single linear layer with ReLU activation, outputting the next-day closing price.

C. Supplementary Architecture Extensions

1) Transformer Extension Models: We extend the primary recurrent benchmark with Transformer-based sequence models adapted to one-step-ahead stock-price forecasting:

- **Informer**: an Informer-oriented encoder with positional encoding, ProbSparse-style attention, and sequence distilling to reduce long-sequence computation [7].
- **Autoformer**: an Autoformer-oriented encoder using moving-average decomposition and an Auto-Correlation-style temporal block to separate trend and seasonal components [8].

For reproducibility, the implementation also keeps lighter Informer-style and Autoformer-style baselines in the codebase. These Lite variants are not the primary Transformer results because they omit the original ProbSparse attention and Auto-Correlation mechanisms.

2) Recurrent Architecture Variants: To test whether the recurrent family can be strengthened without switching to Transformer models, we add ten recurrent variants: LSTM/BiLSTM with pre-normalization, post-normalization, and combined pre+post normalization; CNN-LSTM and CNN-BiLSTM encoders; and LSTM/BiLSTM with temporal attention. The normalization variants apply LayerNorm before and/or after recurrent encoding. The CNN variants apply a one-dimensional convolutional feature extractor before recurrent modeling. The attention variants learn temporal weights over hidden states before the prediction head.

D. Budgeted Hyperparameter Optimization via Genetic Algorithm

To isolate the effect of hyperparameter tuning from architectural choices, we introduce a constrained search procedure based on a Genetic Algorithm (GA). The GA operates on a fixed computational budget and optimizes three hyperparameters using validation loss as the fitness objective:

- **Hidden dimension** (h): searched over $\{32, 64, 128, 256\}$,
- **Learning rate** (α): searched over $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$,
- **Sequence length** (L): searched over $\{10, 20, 30, 60\}$.

Each GA individual encodes one (h, α, L) triplet. The population size is set to 8 and the number of generations is fixed at 5, yielding a total evaluation budget of $8 \times 5 = 40$ configurations per case. Tournament selection (size 3) is used for parent selection, and single-point crossover with a per-gene mutation probability of 0.2 is applied. The hand-picked default configuration used in non-GA cases is ($h = 64, \alpha = 10^{-3}, L = 20$). Configurations tagged as "_ga" in the results tables indicate the GA-optimized setting. This design tests whether budgeted tuning improves accuracy and stability versus fixed defaults, acting as an ablation factor for optimization effort rather than a contribution claim about the search algorithm itself.

For the supplementary Transformer experiments, the code also supports a budgeted random-search mode. This mode uses the same validation objective and a small candidate budget to obtain fast initial estimates before running more expensive GA-style tuning or SHAP analysis.

E. Statistical Validation

Beyond average error reporting, we apply two complementary tests:

- **Wilcoxon signed-rank test**: a nonparametric paired test applied on seed-level RMSE vectors ($n = 5$) to assess whether median performance differences are significant without assuming normality.
- **Diebold-Mariano (DM) test**: applied on aligned forecast-error residuals to evaluate whether one model produces significantly lower prediction errors than another.

The Wilcoxon test provides seed-level robustness evidence, while the DM test assesses time-series-level significance. Agreement between both tests strengthens confidence in reported gains.

V. EXPERIMENTAL SETUP

A. Evaluation Metrics

Primary metrics are MSE, RMSE, MAE, and MAPE on the test partition. We report both point estimates (mean) and seed-wise variability (standard deviation) across five independent runs to characterize stability.

B. Primary Recurrent Benchmark

The main ablation factors are:

- **Model family**: LSTM and BiLSTM
- **Feature regime**: `Without_Macro` (10 features) vs `With_Macro` (16 features)
- **Optimization mode**: default (h=64, α=10^{-3}, L=20) vs GA-optimized ("_ga")

This yields $2 \times 2 \times 2 = 8$ recurrent-model cases per bank for controlled comparison, or 280 recurrent runs across seven banks and five seeds.

C. Supplementary Extension Experiments

Additional experiments evaluate two architecture-extension groups using the same bank and seed protocol. The Transformer extension contains Informer and Autoformer cases under price-only and macro regimes, including fast random-search variants, for 280 runs. The recurrent-variant extension contains normalization, convolutional recurrent, and attention-augmented LSTM/BiLSTM variants, for 1400 runs. These runs are reported as supplementary evidence because paired statistical tests across all model families have not yet been regenerated.

D. Multi-Seed Protocol

Each configuration is executed with five seeds: $\{42, 52, 62, 72, 82\}$. The complete run table stores configuration metadata, per-epoch metrics, and file paths for downstream significance analysis. All cases use identical data splits and seed sets to ensure fairness.

E. Implementation Details

Implementation is based on Python and PyTorch. Training uses the Adam optimizer with ReduceLROnPlateau scheduling and validation-based early stopping. The recurrent baseline uses batch size 32 in the original study; GPU-accelerated Transformer and recurrent-variant sweeps use larger batch configurations where supported by memory. The forecasting horizon is one day. Code and outputs are organized under the project repository, with raw experiment artifacts stored in `code/results_study_full`, `code/results_transformer_study`, and `code/results_recurrent_variants_study`.

F. Cross-Model Fairness Protocol

To maintain fairness, all reported families use identical bank sets, chronological splits, and seed sets. Feature regimes are matched as price-only or macro-augmented. Optimization budgets are kept budgeted rather than exhaustive, so results should be interpreted as controlled experimental evidence rather than a claim of globally optimal hyperparameters.

VI. Results and Discussion

A. Performance Comparison Across Configurations

Table I reports the best original recurrent case per bank based on mean RMSE across five seeds. BiLSTM-based settings are dominant in six out of seven banks. The best configurations are mostly `bilstm_data`, with two bank-specific exceptions: `bilstm_data_ga` for BBTN and `bilstm_macro_ga` for BMRI.

Across all eight recurrent ablation cases, the best overall average RMSE is obtained by `bilstm_data_ga` (680.83), followed closely by `bilstm_data` (681.26), while `lstm_data` is third (741.52). In this dataset, moving from price-only data to macro-augmented data is not uniformly beneficial, and improvements are bank-dependent.

TABLE I
Best-performing recurrent case per bank (mean±std over 5 seeds).

Bank	Best case	RMSE	MAE	MAPE (%)
BBCA	**bilstm_data**	$\mathbf{1881.53 \pm 199.49}$	**1721.55**	**18.36**
BBNI	**bilstm_data**	$\mathbf{335.85 \pm 57.51}$	**252.66**	**4.90**
BBRI	**bilstm_data**	$\mathbf{525.70 \pm 123.24}$	**435.44**	**8.34**
BBTN	**bilstm_data_ga**	$\mathbf{51.44 \pm 16.04}$	**43.48**	**3.23**
BDMN	**lstm_data**	$\mathbf{120.72 \pm 21.92}$	**98.32**	**3.74**
BMRI	**bilstm_macro_ga**	$\mathbf{1388.71 \pm 257.96}$	**1165.79**	**19.09**
BNGA	**bilstm_data**	$\mathbf{244.14 \pm 28.12}$	**192.77**	**11.23**

TABLE II
Best recurrent case vs baseline (`lstm_data`) significance summary.

Bank	Best case	Improve (%)	Wilcoxon p	DM p
BBCA	bilstm_data	8.76	0.4375	2.28×10^{-13}
BBNI	bilstm_data	10.47	**0.0625**	4.84×10^{-11}
BBRI	bilstm_data	28.23	0.3125	$< 10^{-16}$
BBTN	bilstm_data_ga	**35.66**	0.3125	$\mathbf{2.22 \times 10^{-16}}$
BDMN	lstm_data	0.00	1.0000	n/a
BMRI	bilstm_macro_ga	11.18	0.4375	2.23×10^{-11}
BNGA	bilstm_data	4.91	0.3125	4.78×10^{-5}

B. Statistical Significance Analysis

Table II compares each bank's best recurrent case against `lstm_data` as baseline. RMSE gains are positive in six banks, ranging from 4.91% to 35.66%.

C. Global Ranking and Regime Matrix

Table III presents an overall ranking across all banks using mean RMSE, MAE, and MAPE. The top two configurations are both data-only BiLSTM variants, while macro-augmented variants rank below them in the aggregate view.

To directly answer whether macroeconomic feature engineering improves accuracy, Table IV reports the RMSE change from `data` to `macro` per bank. Positive values indicate degradation, and negative values indicate improvement.

This evidence suggests that macro variables are useful only in specific regimes/banks rather than universally. The large RMSE spikes observed for BDMN (+202.86% for LSTM, +126.82% for BiLSTM) are consistent with temporal misalignment between macro indicators and this bank's price dynamics. This pattern is compatible with the broader literature: methods designed for exogenous variables can improve forecasting when the external signals are informative and properly aligned [1], [9], and multiple IEEE studies also report gains from macro-integrated forecasting pipelines [2], [10], [11].

D. XAI Analysis with SHAP

We generated SHAP outputs for all 280 original recurrent checkpoints (7 banks $\times$ 8 cases $\times$ 5 seeds). This produced 3,640 feature-attribution rows over 16 unique features. Using global top-1 frequency, the most frequently dominant drivers are *Low Price* (39 runs), *Bid Price* (31), *Price Change 1 Day Percent* (28), and *Price to Sales Ratio* (28). The pattern indicates that short-horizon market microstructure and valuation factors dominate the model explanations more often than macro indicators.

TABLE III
OVERALL RECURRENT RANKING ACROSS ALL BANKS (LOWER IS BETTER).

Case	RMSE	MAE	MAPE (%)
bilstm_data_ga	**680.83**	591.04	**10.65**
bilstm_data	681.26	**589.76**	10.74
lstm_data	741.52	644.21	11.44
lstm_data_ga	781.95	688.86	12.66
bilstm_macro_ga	806.40	722.52	13.76
bilstm_macro	814.04	720.31	13.14
lstm_macro	851.89	759.88	13.96
lstm_macro_ga	938.15	842.31	17.48

TABLE IV
RMSE CHANGE (%) FROM `DATA` TO `MACRO`.

Bank	LSTM delta (%)	BiLSTM delta (%)
BBCA	+20.34	+27.87
BBNI	+30.70	+39.21
BBRI	-1.27	+23.19
BBTN	**-5.04**	**-17.72**
BDMN	+202.86	+126.82
BMRI	-1.76	-4.85
BNGA	+13.25	+15.45
Mean	+37.01	+29.99

Because a small number of runs showed extreme SHAP magnitudes (maximum $\approx 1.51 \times 10^{13}$), we also report a robust global ranking using a 99th-percentile cap. Under this robust aggregation, the top features remain stable: *Low Price* (0.00465), *Bid Price* (0.00428), *Price Change 1 Day Percent* (0.00407), *High Price* (0.00394), and *Moving Average 20 Day* (0.00383). Hence, the main interpretation does not depend on a few outlier explanations.

E. Supplementary Transformer Extension

Table VI summarizes the strongest global case from each completed experiment family. As an extension to the recurrent benchmark, the Transformer sweep shows the strongest aggregate performance. The best Transformer case, `informer_data`, reduces mean RMSE from 680.83 in the best original recurrent case to 401.02.

The best Transformer case outperforms the best original recurrent case in every bank-level comparison. The average best-per-bank improvements are 38.31% for RMSE, 42.17% for MAE, and 41.95% for MAPE. Table VII reports the RMSE comparison in detail. These results indicate strong potential for Informer and Autoformer modeling, but they are reported as extension evidence because paired statistical tests against the recurrent winners have not yet been regenerated.

TABLE V
GLOBAL SHAP SUMMARY (ROBUST AGGREGATE OVER ALL 280 RECURRENT RUNS).

Feature	Mean —SHAP— (cap99)	Top-1 count
Low Price	**0.00465**	**39**
Bid Price	0.00428	31
Price Change 1 Day Percent	0.00407	28
High Price	0.00394	25
Moving Average 20 Day	0.00383	25
Price to Book Ratio	0.00383	25
Price Earnings Ratio	0.00365	20
Price to Sales Ratio	0.00362	**28**

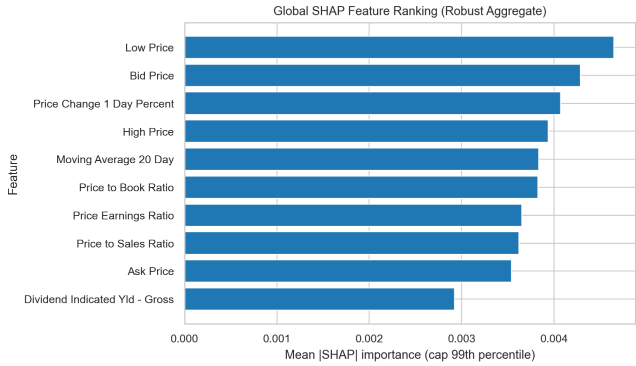

Fig. 2. Global SHAP ranking using robust mean importance (99th-percentile capped).

F. Supplementary Recurrent-Variant Extension

The recurrent-variant sweep contains 1400 runs over normalization, convolutional recurrent, and attention-augmented LSTM/BiLSTM cases. The strongest global variant is `cnn_bilstm_data_ga`, with RMSE 754.96, MAE 668.21, and MAPE 12.35%. The top three recurrent variants are `cnn_bilstm_data_ga`, `cnn_bilstm_data`, and `cnn_lstm_data`, indicating that convolutional preprocessing is the most competitive recurrent-variant subgroup. Table VIII lists the best recurrent variant for each bank.

However, recurrent variants do not improve the original recurrent benchmark on average. Their average best-per-bank differences relative to the original recurrent winners are -10.18% RMSE, -7.90% MAE, and -7.41% MAPE, where negative values indicate worse performance. The only bank-level exception is BDMN, where `cnn_bilstm_data_ga` slightly improves RMSE over `lstm_data` (116.73 vs. 120.72). Recurrent variants are therefore useful as robustness and architecture-ablation evidence, but they do not replace the original BiLSTM baseline in this study.

G. Discussion and Threats to Validity

Three validity points are important. First, the chronological split is fixed; different split boundaries may change relative ranking. Second, seed-level Wilcoxon tests are underpowered with $n = 5$, which explains why several improvements with very small DM p-values are not mirrored by Wilcoxon significance. Third, optimization benefits are heterogeneous across banks, indicating risk of overfitting when tuning is not controlled.

The extension experiments add two additional caveats. Transformer and original recurrent runs have SHAP out-

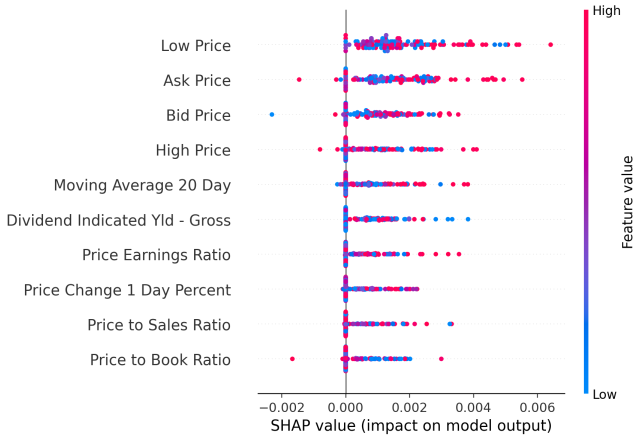

Fig. 3. SHAP beeswarm plot for the best-performing recurrent checkpoint (BBTN, BiLSTM_Data_GA, seed 62).

TABLE VI
GLOBAL BEST CASE BY EXPERIMENT FAMILY.

Family	Best case	RMSE	MAE	MAPE (%)
Original recurrent	`bilstm_data_ga`	680.83	591.04	10.65
Recurrent variants	`cnn_bilstm_data_ga`	754.96	668.21	12.35
Transformer	**`informer_data`**	**401.02**	**325.11**	**5.94**

puts, but recurrent variants were executed without SHAP to control compute cost. In addition, the Transformer and recurrent-variant results currently provide multi-seed aggregate evidence, while the strongest inferential validation remains the original recurrent Wilcoxon and DM analysis. Future work should regenerate paired statistical tests across the best recurrent, Transformer, and recurrent-variant cases before making stronger cross-family significance claims.

VII. Conclusion

This paper presents a reproducible framework for forecasting Indonesian bank stock prices using LSTM-family models, macroeconomic features, multi-seed evaluation, explainability outputs, and statistical significance testing.

Empirical results show that BiLSTM-based settings are generally stronger than LSTM across the evaluated banks, with best-case RMSE gains over the `lstm_data` baseline reaching up to 35.66%. Diebold-Mariano tests confirm significant forecast-error reduction in six banks, while Wilcoxon results are less conclusive due to limited seed count. The evidence also indicates that macro features and optimization are not uniformly beneficial, motivating bank-specific model selection rather than one-size-fits-all deployment.

The supplementary architecture experiments provide a stronger extension path. Informer/Autoformer Transformer runs achieve the best aggregate errors, with `informer_data` reaching RMSE 401.02, MAE 325.11, and MAPE 5.94%. Recurrent variants based on normalization, convolution, and attention broaden the ablation space, but they do not improve the original recurrent baseline on average. These results suggest that Transformer models are the most promising next stage, while recurrent variants are better interpreted as robustness experiments.

TABLE VII
BEST TRANSFORMER IMPROVEMENT OVER BEST ORIGINAL RECURRENT CASE.

Bank	Best recurrent	Rec. RMSE	Best Transformer	Tr. RMSE	Improve (%)
BBCA	`bilstm_data`	1881.53	`autoformer_macro`	545.93	70.98
BBNI	`bilstm_data`	335.85	`informer_data`	219.66	34.60
BBRI	`bilstm_data`	525.70	`informer_macro`	379.21	27.87
BBTN	`bilstm_data_ga`	51.44	`informer_data_random`	43.01	16.40
BDMN	`lstm_data`	120.72	`informer_data`	76.82	36.37
BMRI	`bilstm_macro_ga`	1388.71	`informer_data_random`	885.21	36.26
BNGA	`bilstm_data`	244.14	`informer_data`	132.56	45.70
Mean	–	–	–	–	**38.31**

TABLE VIII
BEST RECURRENT-VARIANT CASE PER BANK.

Bank	Best variant	RMSE	MAE	MAPE (%)	RMSE vs rec. (%)
BBCA	`bilstm_attention_data`	2004.10	1830.50	19.51	-6.51
BBNI	`cnn_bilstm_data`	362.88	263.19	5.08	-8.05
BBRI	`bilstm_attention_data`	580.65	454.46	8.59	-10.45
BBTN	`bilstm_attention_data`	70.84	58.99	4.35	-37.71
BDMN	`cnn_bilstm_data_ga`	116.73	89.56	3.40	**3.30**
BMRI	`cnn_bilstm_macro`	1533.72	1299.75	21.29	-10.44
BNGA	`cnn_bilstm_data_ga`	247.60	196.96	11.48	-1.41

Several limitations should be noted. First, the evaluation is restricted to seven Indonesian banking stocks, which constrains generalizability to other markets and sectors. Second, the five-seed protocol limits statistical power for nonparametric tests. Third, Transformer and recurrent-variant statistical tests should be regenerated from raw prediction outputs to match the original recurrent validation depth. Fourth, recurrent variants currently lack SHAP outputs because the 1400-run sweep was executed without SHAP for compute-cost control.

Immediate future work includes: (i) extending validation to walk-forward protocols, (ii) increasing seed count for stronger nonparametric testing power, (iii) ablation on feature lag to investigate whether temporal misalignment explains macro feature failures, and (iv) paired statistical validation and targeted SHAP analysis across the best recurrent, Transformer, and recurrent-variant cases.

References

[1] Y. Wang, H. Wu, J. Dong, G. Qin, H. Zhang, Y. Liu, Y. Qiu, J. Wang, and M. Long, "Timexer: Empowering transformers for time series forecasting with exogenous variables," *arXiv preprint arXiv:2402.19072*, 2024. [Online]. Available: https://arxiv.org/abs/2402.19072

[2] S. F. Adyatma and A. Alamsyah, "The indonesia stock exchange composite prediction based on macroeconomic indicators using arima, lstm, and ann," in *2022 8th International Conference on Science and Technology (ICST)*, 2022.

[3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[4] Y. Yang and J. Xu, "Stock index prediction based on cnn-bilstm-attention," in *2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE)*, 2024.

[5] Y. Saleh, M. Pradana, and B. R. Kartawinata, "Comparative forecasting of indonesian blue-chip stocks using arima, ann and lstm models," in *2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC)*, 2025.

[6] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *arXiv preprint arXiv:1912.09363*, 2019. [Online]. Available: https://arxiv.org/abs/1912.09363

[7] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," *arXiv preprint arXiv:2012.07436*, 2020. [Online]. Available: https://arxiv.org/abs/2012.07436

[8] H. Wu, J. Xu, J. Wang, and M. Long, “Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting,” *arXiv preprint arXiv:2106.13008*, 2021. [Online]. Available: https://arxiv.org/abs/2106.13008

[9] X. Qiu, Y. Zhu, Z. Li, X. Wu, B. Yang, and J. Hu, “Dag: A dual correlation network for time series forecasting with exogenous variables,” *arXiv preprint arXiv:2509.14933*, 2025. [Online]. Available: https://arxiv.org/abs/2509.14933

[10] S. S. Begum, N. S. Fatima, and M. Abbas, “Enhancing stock market prediction with an lstm-attention model integrating macroeconomic and sentiment features,” in *2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 2024.

[11] J. Dechjijaruwate and C. Pisarn, “Forecasting stock prices of commercial banks in the stock exchange of thailand using machine learning,” in *2025 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, 2025.

[12] C. M. Shih and K. C. Yang, “Using lightgbm + shap to analyze the impact of factors on stock market investment,” in *2024 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 2024.

[13] M. H. Begum, F. M. A. Rahman, M. A. K. Azad, C. R. Debi, K. M. M. Uddin, and A. K. M. Masum, “Diverse minds think alike: A shap-guided rnn for stock price forecasting with smart sma optimization,” in *2025 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)*, 2025.

[14] Y. Gupta S, P. Naresh, C. Nagesh, P. Praveen, N. Kirankumar, and K. Surendra, “An interpretable framework for stock market forecasting using long short-term memory (lstm) networks with shap - explainable ai (xai) method,” in *2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, 2025.

[15] H. Tanveer, S. Arshad, H. Ameer, and S. Latif, “Interpretability in financial forecasting: The role of explainable ai in stock market,” in *2024 14th International Conference on Software Technology and Engineering (ICSTE)*, 2024.

[16] H. Tanveer and S. Latif, “Decoding stock market predictions: Insights from explainable ai using layer-wise relevance propagation,” in *2024 26th International Multi-Topic Conference (INMIC)*, 2024.

[17] J. Jagannathan and V. V. S. K. Goditi, “Integration of a hybrid model with xai for stock price forecasting: A dashboard-driven approach,” in *2024 International Conference on IT Innovation and Knowledge Discovery (ITIKD)*, 2025.

[18] S. Ayyappan, A. A. Gadgil, R. Gupta, H. B. Furaiji, S. Najah, and J. A. Jasim, “Time series forecasting framework using explainable deep learning for stock price prediction,” in *2025 International Conference on Recent Innovation in Science Engineering and Technology (ICRISET)*, 2025.